tors have been asked to inspect and report on their fleet within a week "to ensure continued airworthiness and safety of operations". DGCA chief Faiz Ahmed Kidwai did not respond to calls from INDIA TODAY.

Fuel control switches are critical components used to manage the flow of fuel to the engines. The AAIB report found that, moments after takeoff, both engines shut down—after the fuel control switches had moved from the 'RUN' to the 'CUTOFF' position "one after another with a time gap of one second". One of the pilots attempted to turn them back on, but the aircraft had already lost too much altitude and crashed into nearby buildings.

The FAA's Special Airworthiness Information Bulletin in December 2018 had warned that these switches on Boeing aircraft could be inadvertently moved, leading to "in-flight engine shutdown", if installed with the locking feature disengaged. Air India admitted to investigators that it had never inspected the switches on the ill-fated aircraft, citing the FAA advisory's non-mandatory nature.

PILOT OR COMPUTER?

So, the big question: who flipped the fuel switches and why? The report, paraphrasing the Cockpit Voice Recorder (CVR) exchange, notes: "One of the pilots is heard asking the other why did he cutoff [sic]. The other pilot responded that he did not do so." It does not, however, identify who said which part.

Since the switches are designed to require manual unlocking before operation, some aviation experts find it implausible that a pilot could accidentally flip both within a second—suggesting a technical glitch. Sam Thomas, president of the Airline Pilots Association of India, calls the report "biased" against the crew. He claims the truth will emerge only when the entire CVR transcript, not just paraphrased excerpts, are made public.

Others point to the complex interplay of systems aboard the Dreamliner, highlighting potential flaws that need not entail human error. A 2020–21 bulletin from GE, issued via the FAA, warned that on certain GEnx-2B engines, tiny solder balls beneath a microprocessor could crack over time. This degradation in the electronic engine control (EEC) units, the bulletin noted, could lead to a mid-flight "loss of thrust control". Critics cite historical issues with Boeing 787's computer-triggered thrust reduction, suggesting a possible failure of the Thrust Control Malfunction Accommodation system, which works in tandem with EEC, to help differentiate whether the plane is airborne or on the ground.

Then there are those who suspect intentional human action. Captain Steve Scheibner, a US-based commercial pilot and aviation commentator, asks what could drive a pilot to manually cut off both fuel supplies and then deny it. He points to the inclusion of an "aviation psychologist" on the AAIB's investigation team as a possible indicator that mental health is being explored as a cause. For those of his ilk, the incident echoes the Germanwings Flight 9525 crash of 2015, where the co-pilot—previously treated for "suicidal tendencies"—deliberately crashed an Airbus A320 into the French Alps, killing 150 people.



Who Flipped the Switch?

- **Seconds after takeoff, both fuel control switches moved from 'RUN' to 'CUTOFF', shutting down both engines**
- **The switches require deliberate unlocking, casting doubt on both being flipped accidentally within a second**
- **Experts remain divided—was it a system fault, a procedural response to engine failure, or a deliberate act?**

INTERVENTION OR ERROR?

But there's a competing theory: that the switches were manually moved because the engines had already failed, and not the other way around. Aviation safety expert Captain Mohan Ranganathan describes the event as "human intervention" rather than human error. He speculates that the pilots may have been trying to follow Boeing's standard procedure for dual-engine failure: switch both fuel levers to 'CUTOFF', then back to 'RUN' within a second. According to Ranganathan, the flight's data logs, available in the preliminary report, suggest the crew was attempting this sequence. "The next step is to deploy the Ram Air Turbine (RAT) by pressing and holding the RAT switch for one second," he adds. (RAT provides emergency power to the aircraft in case of total engine failure.) However, they may not have been able to complete the final requirement—accelerate to 270 knots; the maximum recorded speed was 180 knots. Sanjeev Kapoor, a retired air marshal, reinforces this possibility, noting that a 'Mayday' was sent and the RAT was deployed.

The final investigation report is likely still months away and, with it, definitive answers. For now, the AAIB's inquiry remains a complex, multi-agency global effort, also involving Boeing, GE and the US National Transportation Safety Board. Air India CEO Campbell Wilson has cautioned against premature conclusions, emphasising that the preliminary findings did not point to any mechanical or maintenance faults in the aircraft. However, for the families of the 260 victims—241 on board and 19 on the ground—such assurances offer little solace. In the end, the answers unearthed from this tragedy may do more than explain what went wrong—they may help ensure that such a disaster never happens again. ■